

APPLEJAM ENGINE

LET'S GO, MOVE!

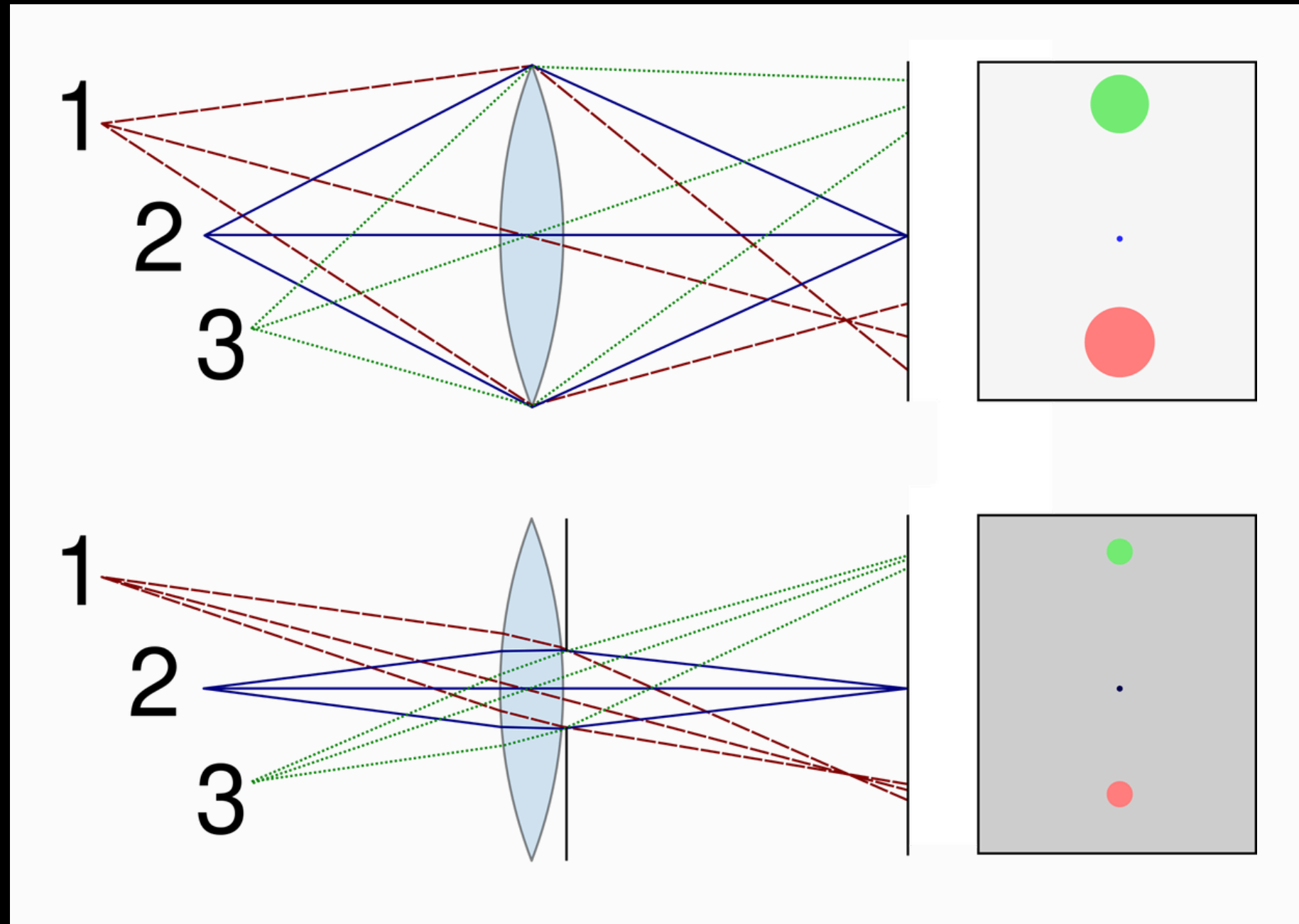
JUNGLE GAME TECH LAB 3 W13
TEAM2 김태형 김형도 정찬일 이상훈

목차

- Depth of Field
- Degree of Freedom
- PhysX
- Vehicle
- Cloth

Depth of Field

피사계 심도

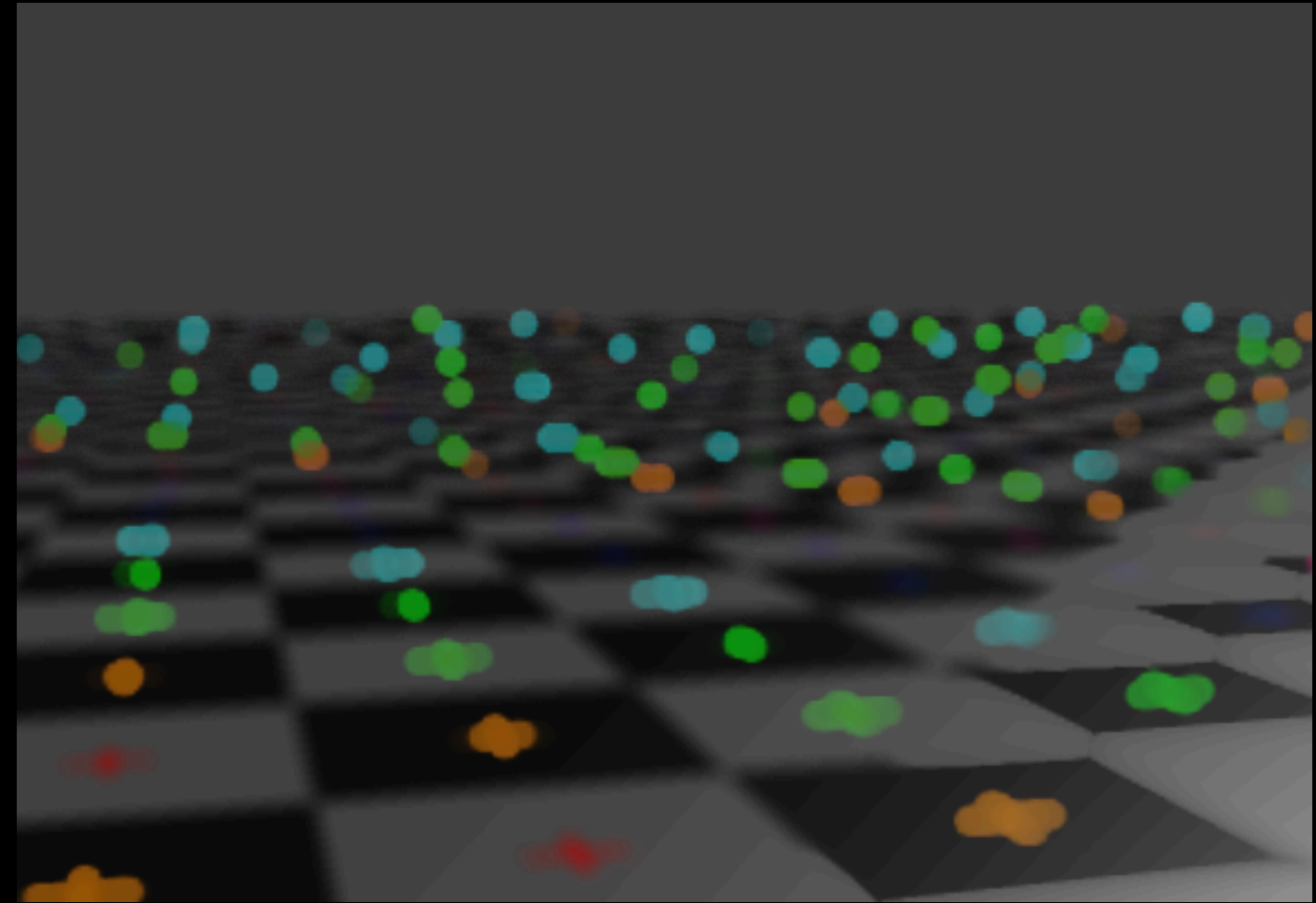
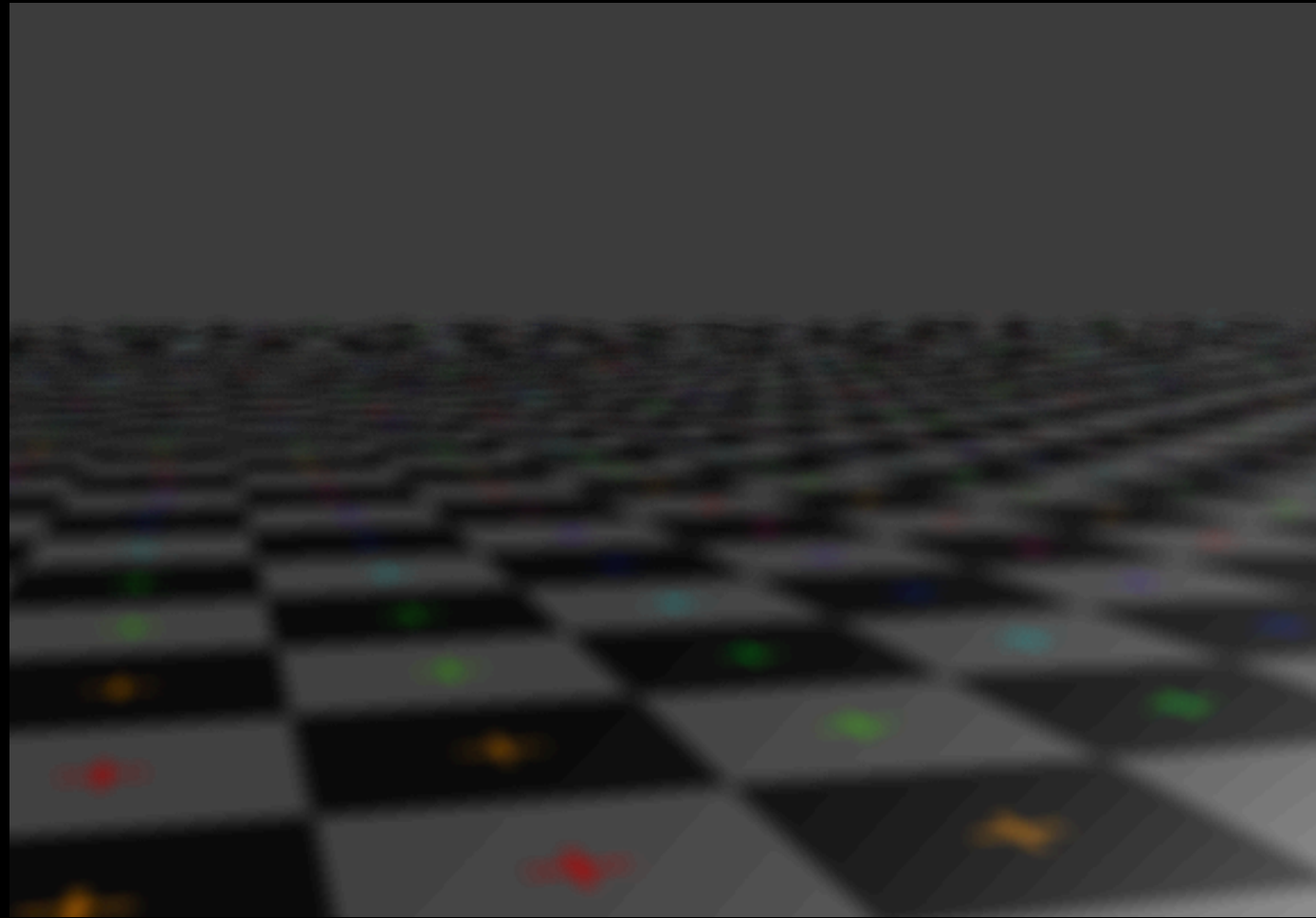


조리개 값이 클 수록
초점이 맞지 않을 때
빛이 더 크게 퍼진다

착란원(Circle of confusion, CoC)
이 커진다

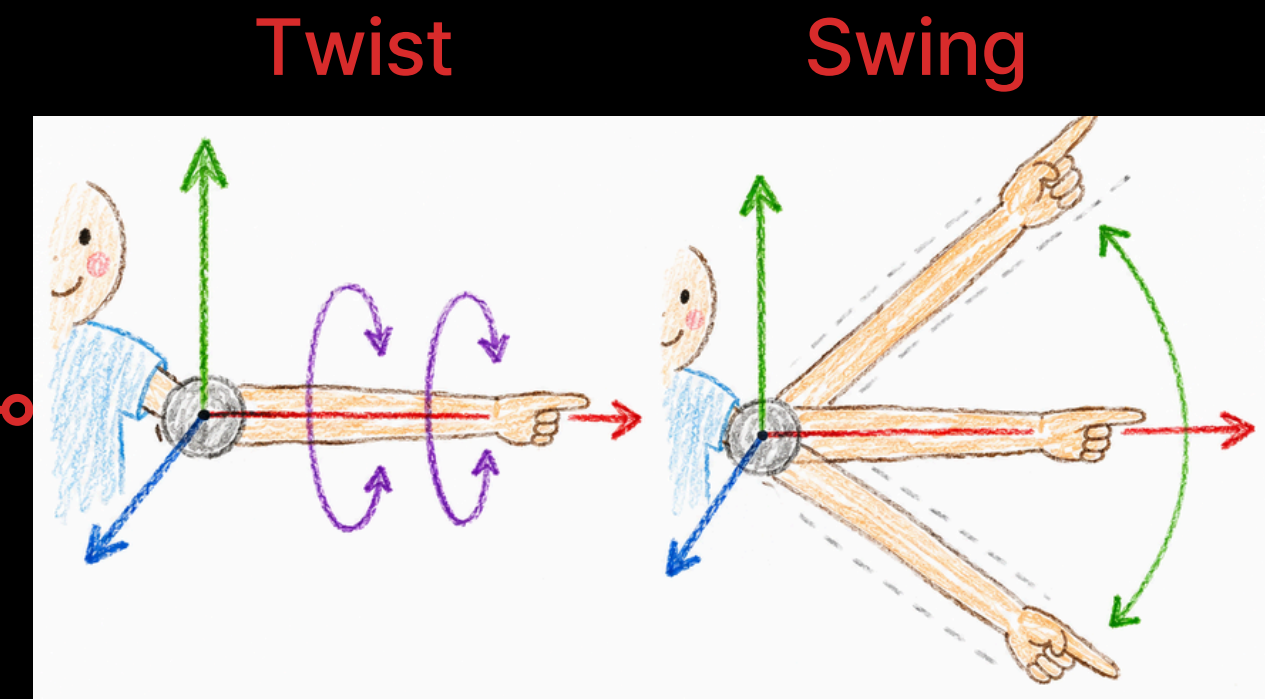
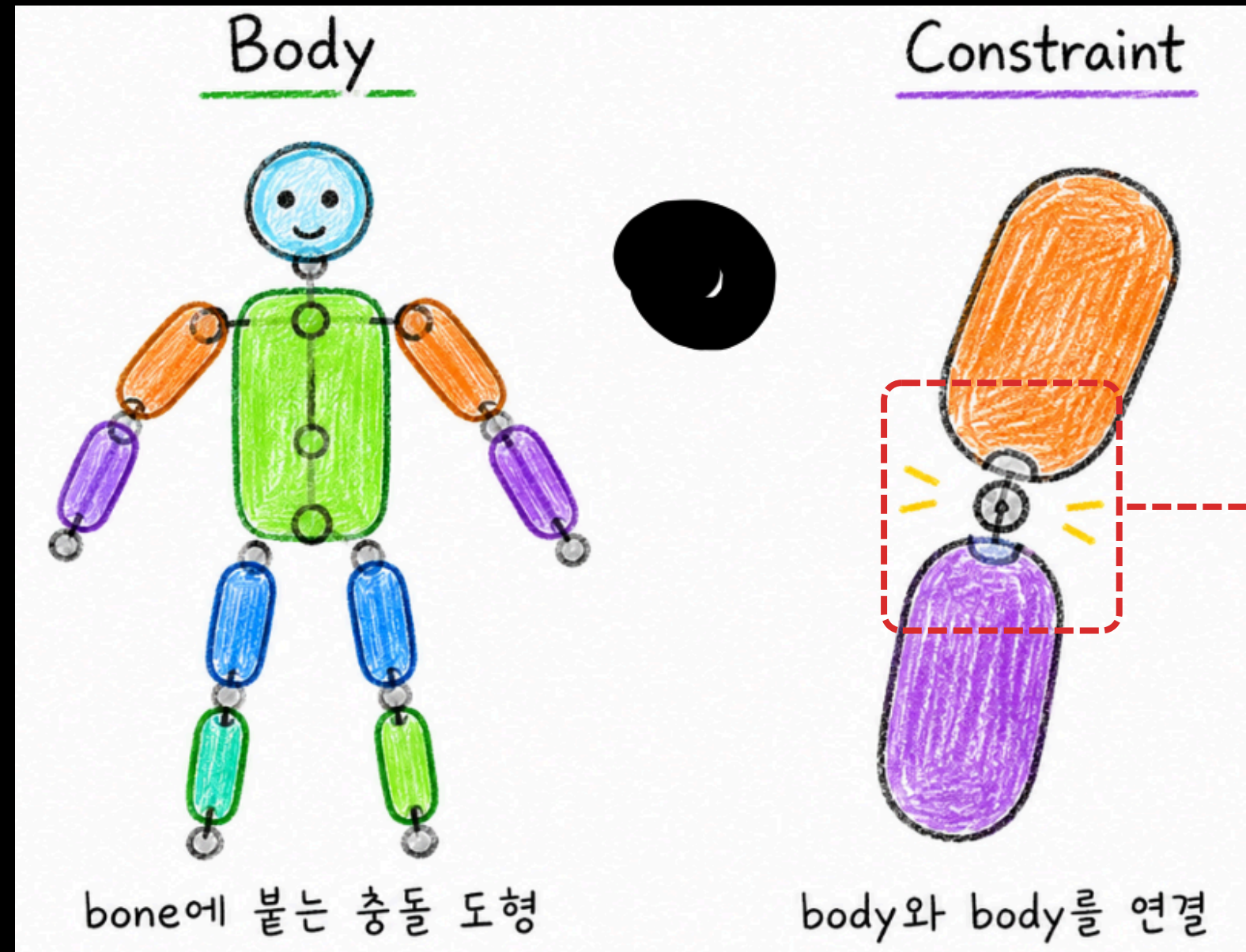
더 뿌옇게 보인다

Depth of Field

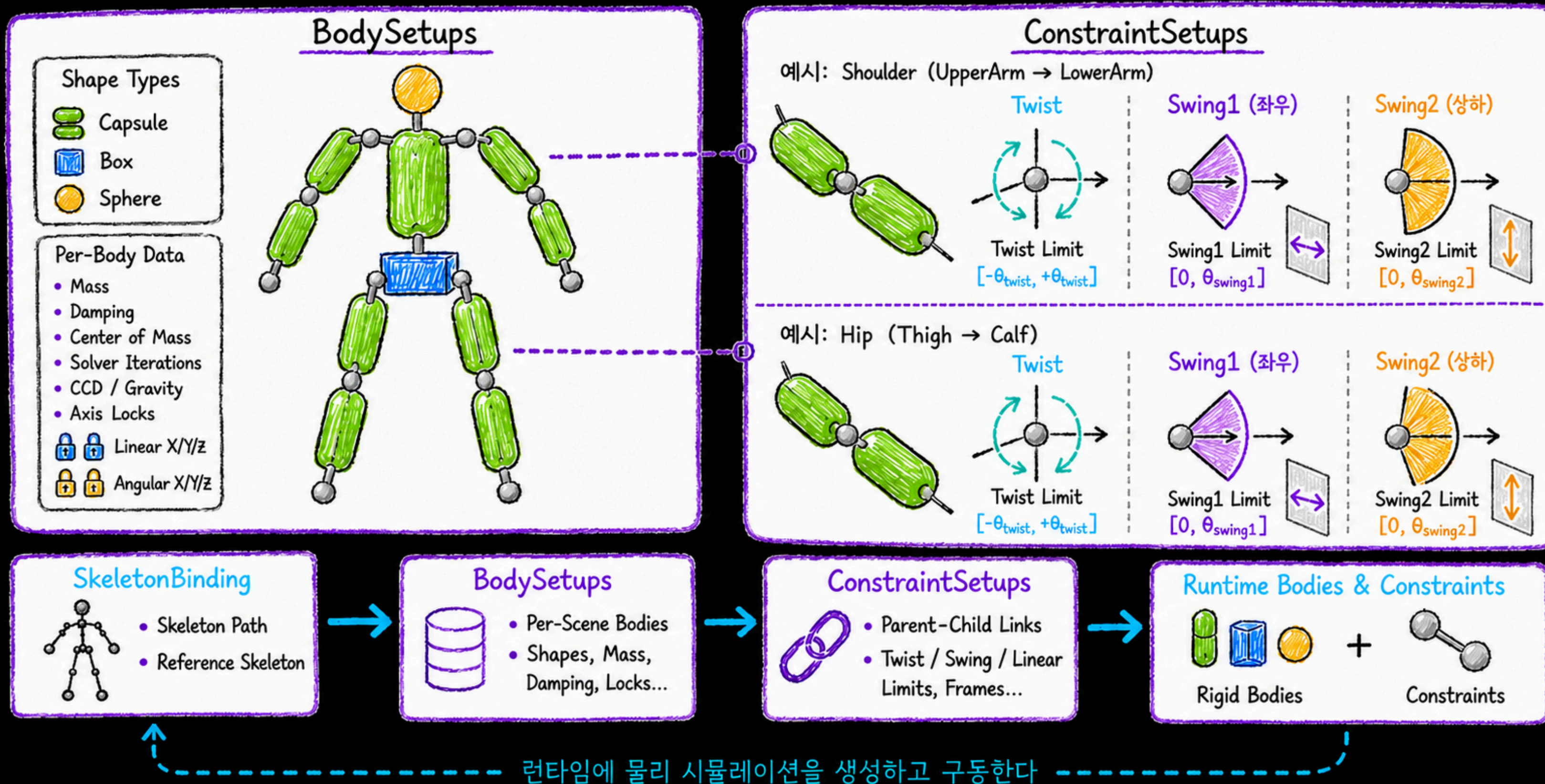


착란원 크기와 밝기가 일정 수준 이상인 경우
스프라이트를 합성하여 보케 효과 연출

Degree of Freedom

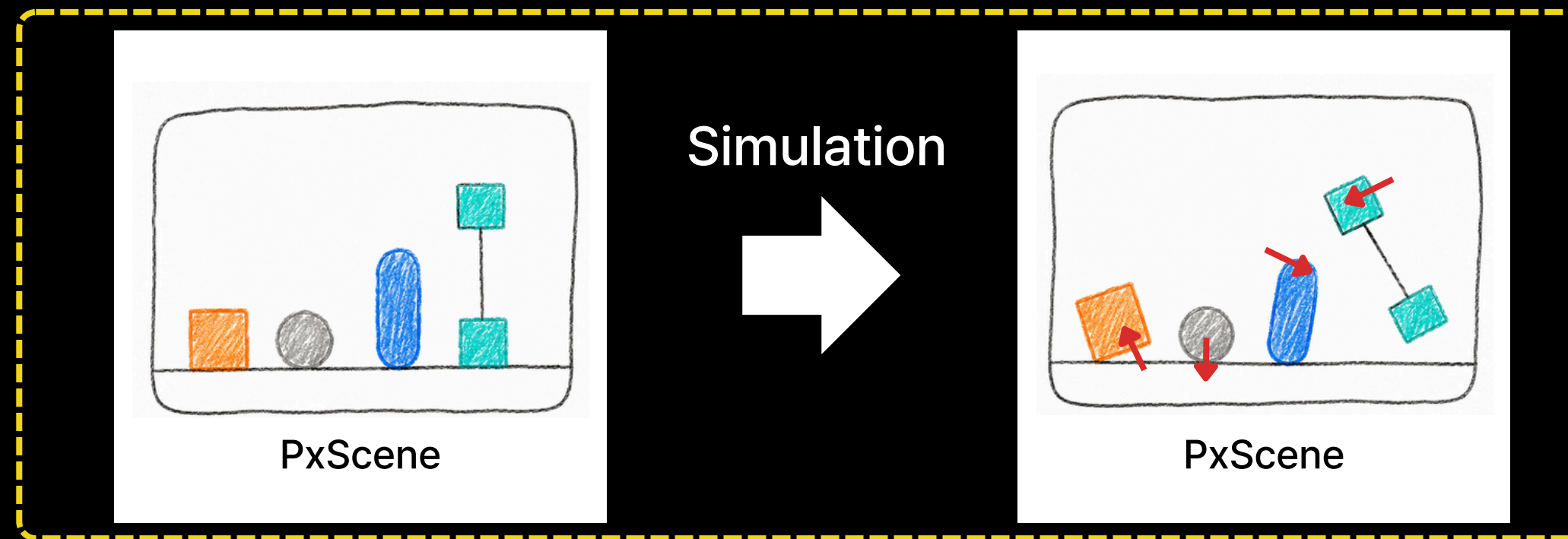


UPhysicsAsset



PhysX

PhysX



Copy
PxRigidActor, PxShape, PxJoint

Fetch

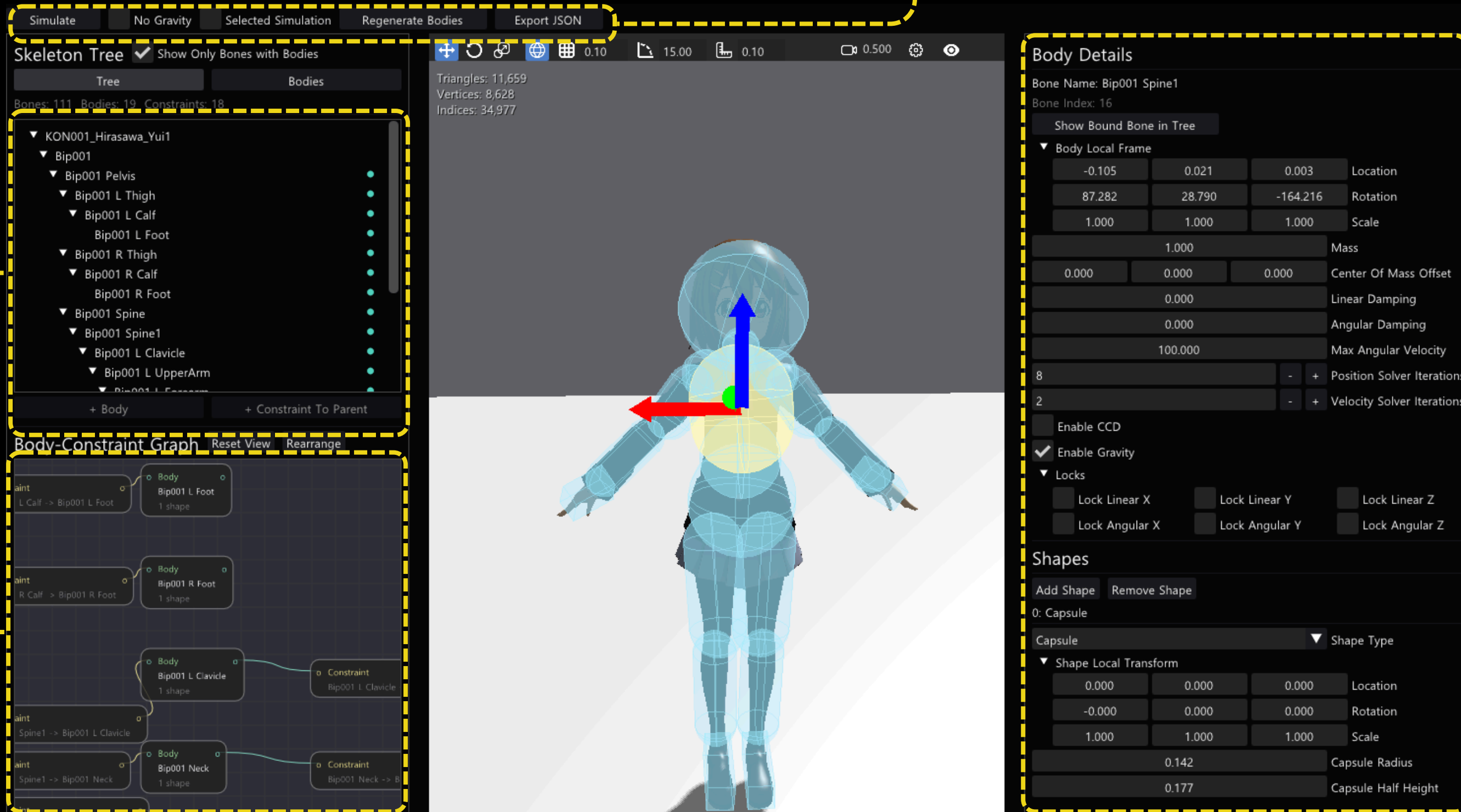
Engine



Physics Editor

Simulation 및 Body 자동 생성 기능

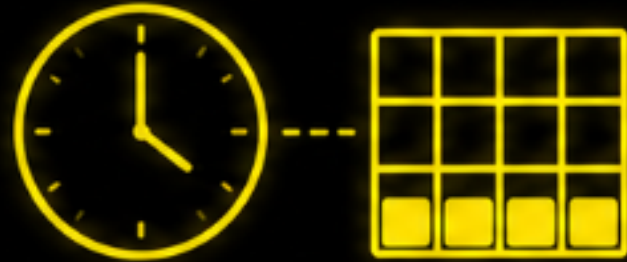
Skeleton Tree



Body ,
Constraint
관련 설정

Delta Time

1. Fixed-Step Accumulator



가변 DeltaTime을 누적
해 FixedDt 단위로 처리
한다.

2. Physics Substep



Fixed Dt를 더 작은
Substep으로 나눠
안정적으로
simulate한다.

3. Accumulator Drop



MaxSubsteps를 넘으면
남은 누적 시간을 버려
연쇄 지연을 막는다.

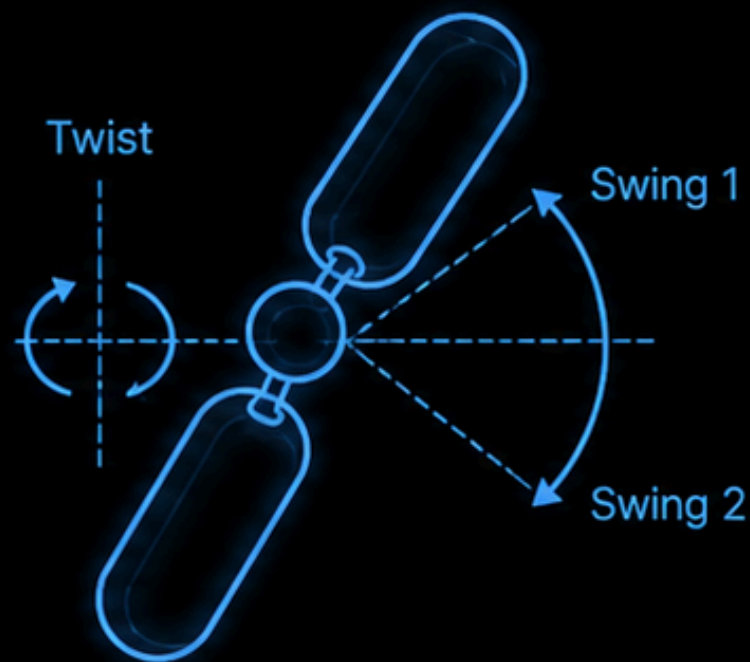
Ragdoll

1. Skeletal Body



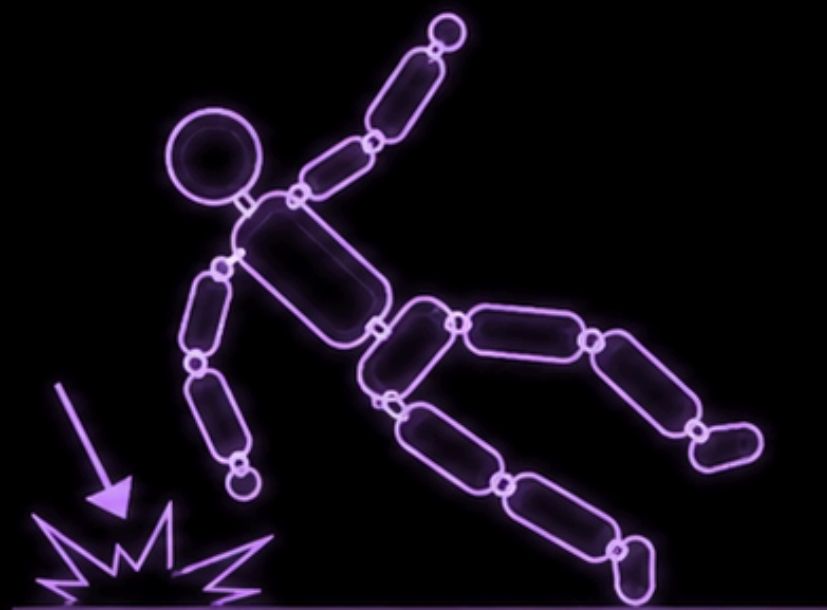
뼈대의 각 bone에
물리 body를 붙여
충돌과 질량을 만든다.

2. Constraint Link



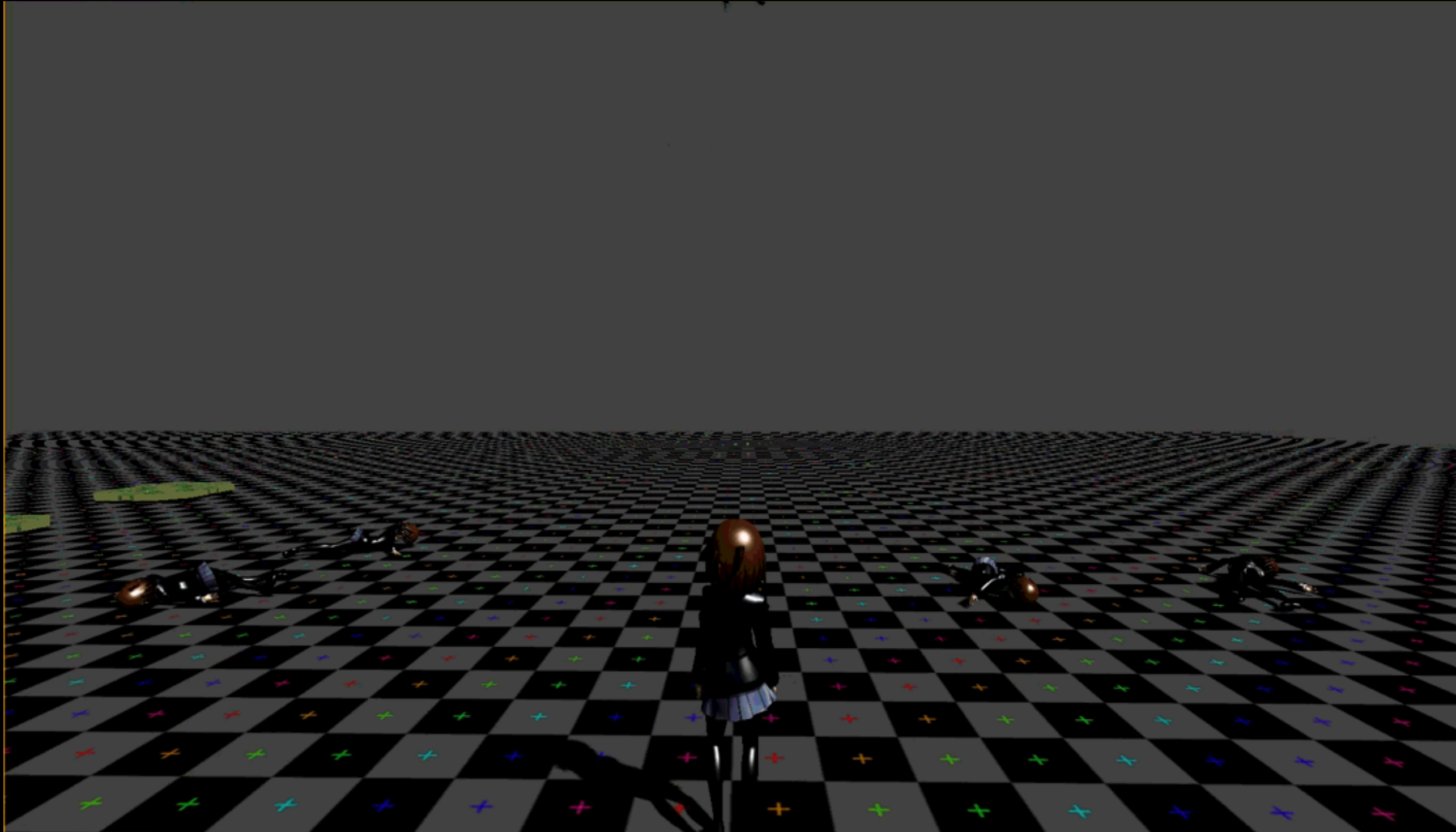
body와 body를
constraint로 연결해
관절 움직임을 제한한다.

3. Physics Reaction



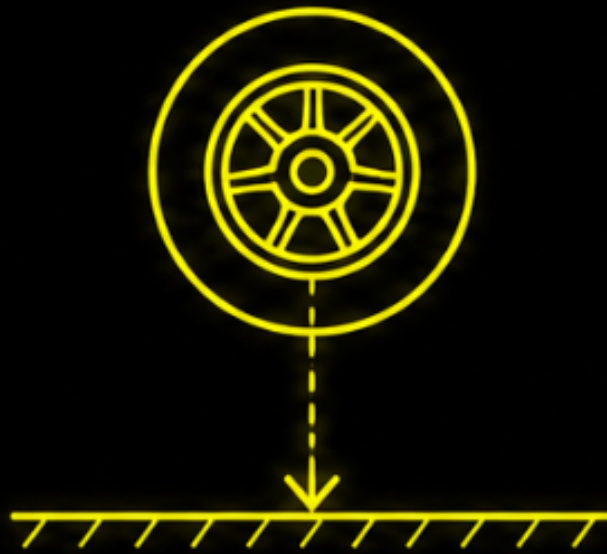
애니메이션 대신
물리 계산으로 넘어가
넘어짐을 자연스럽게 만든다.

Ragdoll

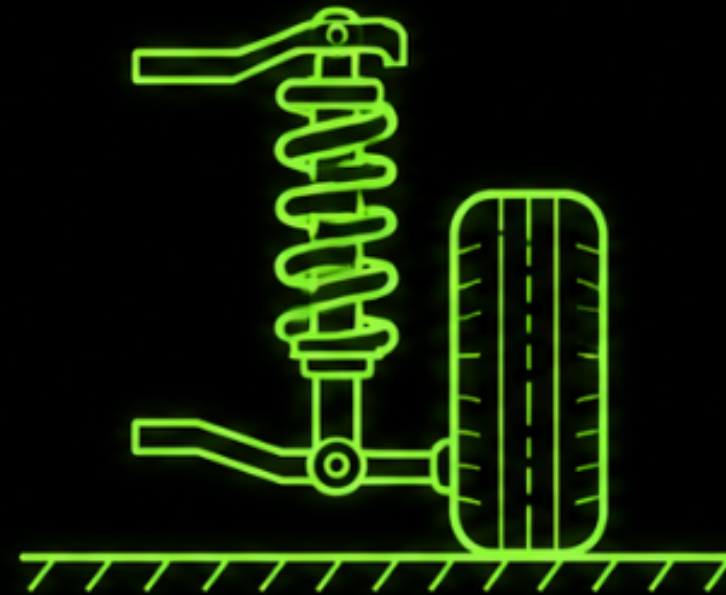


Vehicle

Raycast



Suspension

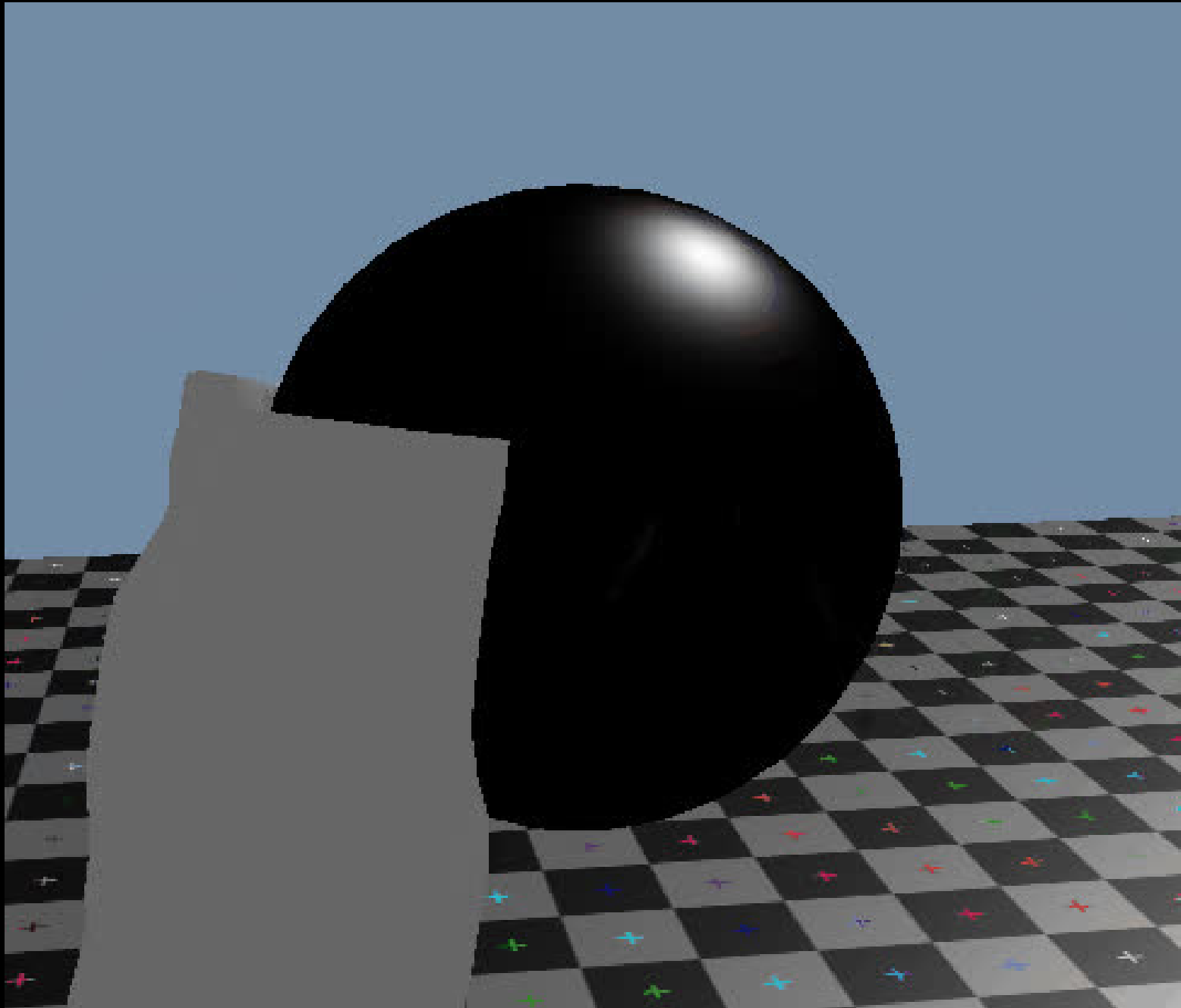


Tire Friction



Raycast로 접지를 판단하고
Suspension으로 차체를 지지하며 Tire Friction으로 힘을 전달한다.

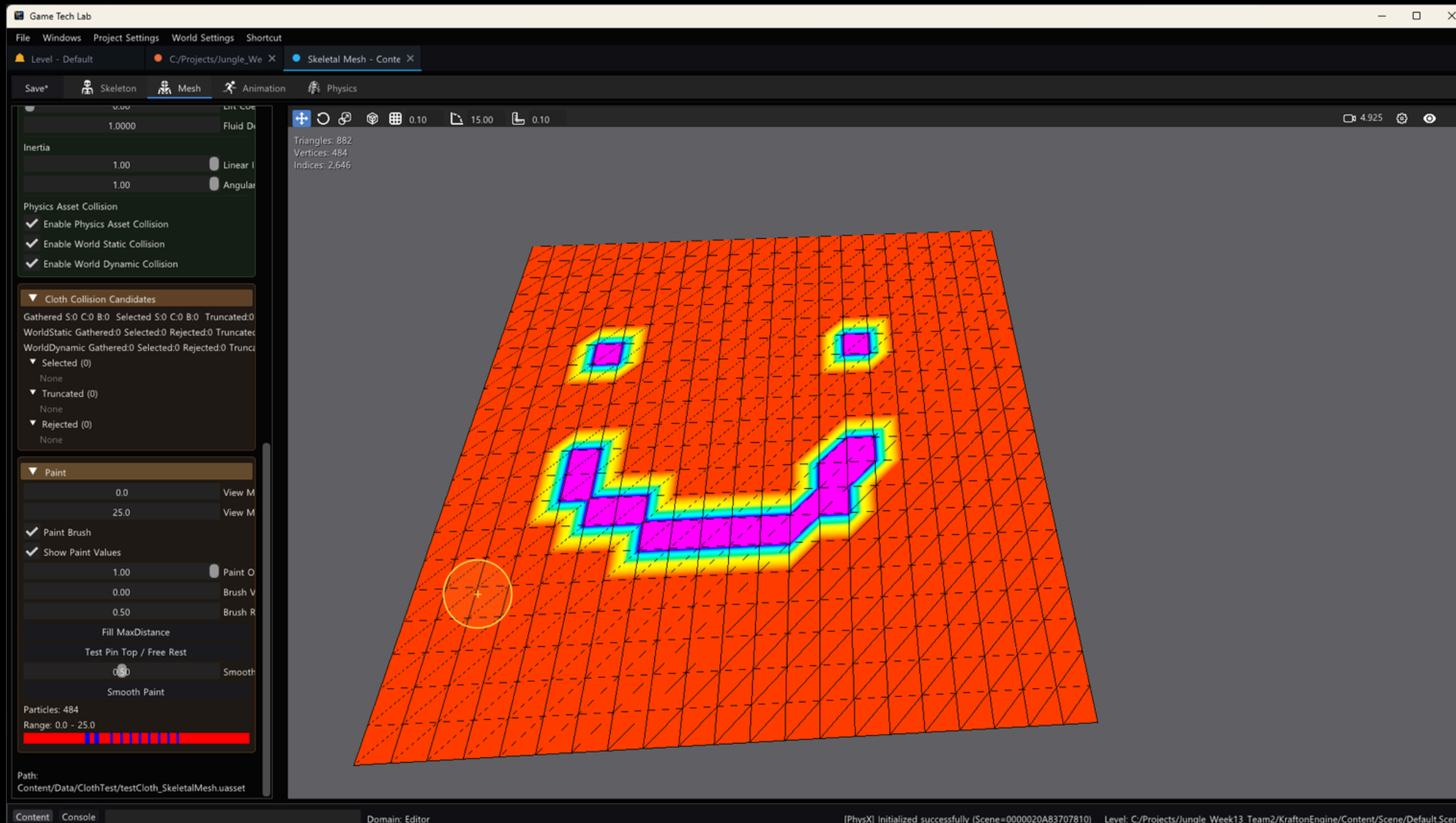
Cloth Simulation



NvCloth를 사용한 Cloth
Simulation 기능 추가
Skeletal Mesh Section 단위로
적용 가능

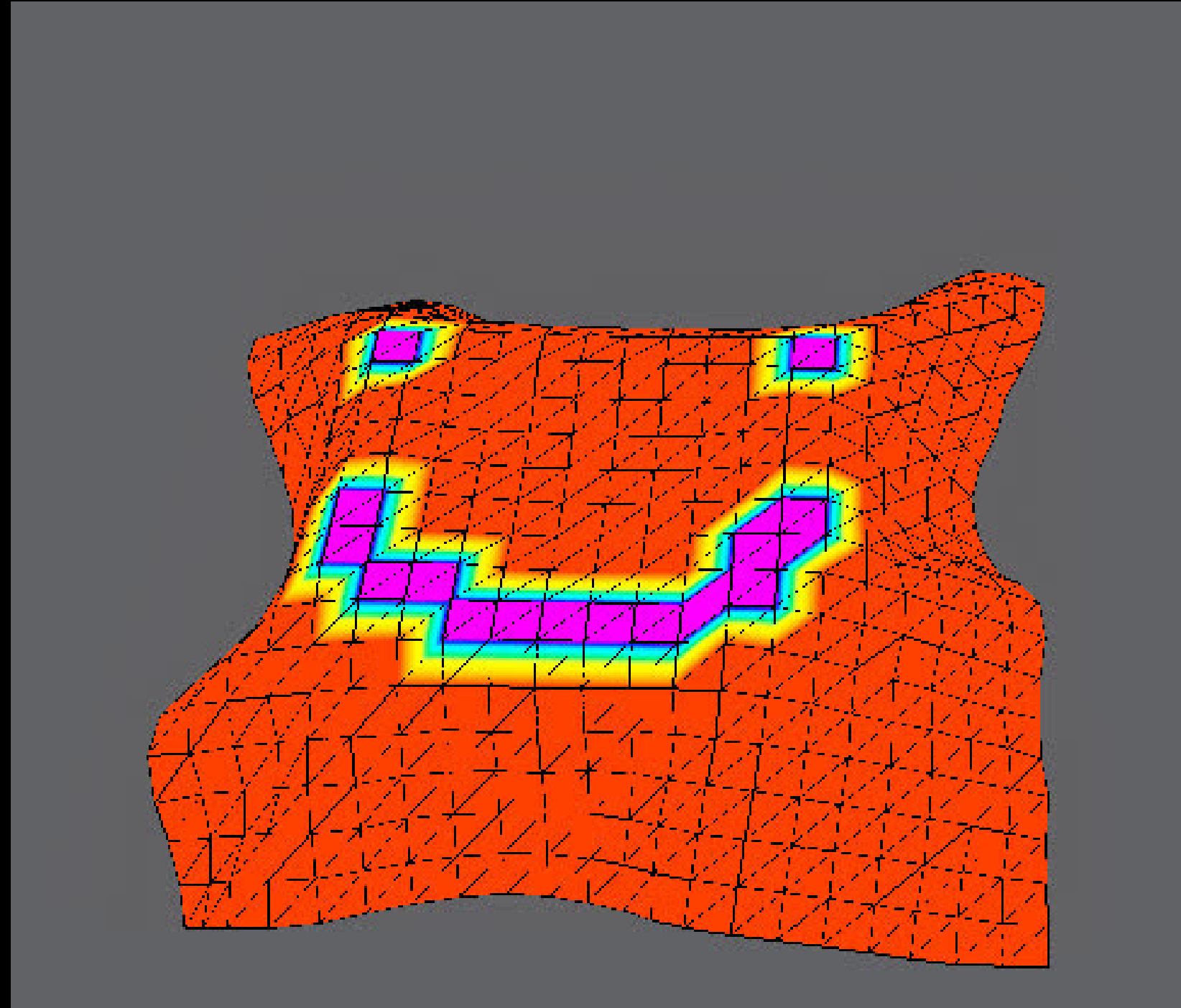
Cloth Simulation

Cloth Paint 기능



시뮬레이션으로 얼마나
자유롭게 움직일 수 있는지
정점별로 Max Distance
설정 가능

Cloth Simulation





시연